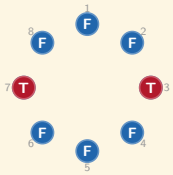


The Vote-Left Equilibrium: A Deterministic Coordination Strategy for the Faithful in The Traitors

Vince Knight, Cardiff

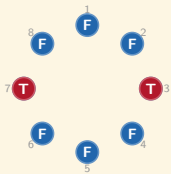
A

Roles



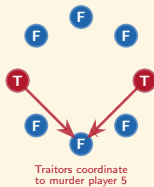
A

Roles



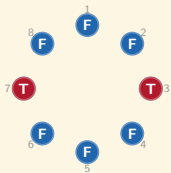
B

Night: Murder

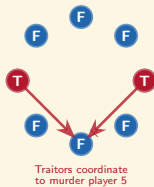


A

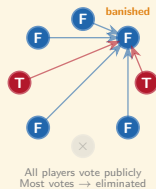
Roles

**B**

Night: Murder

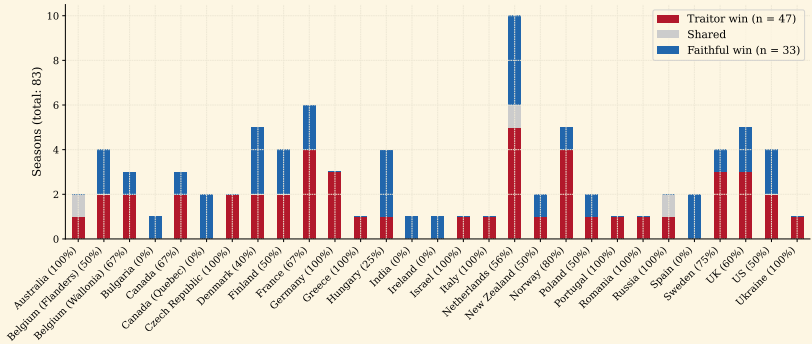
**C**

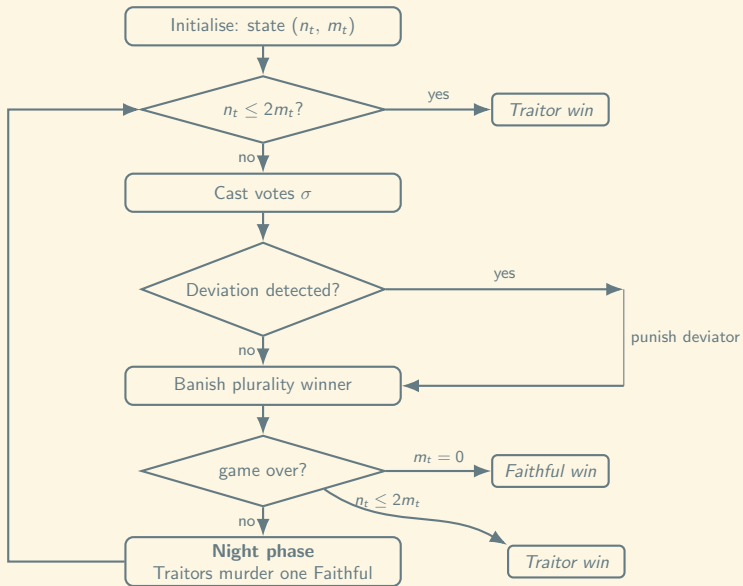
Day: Banishment



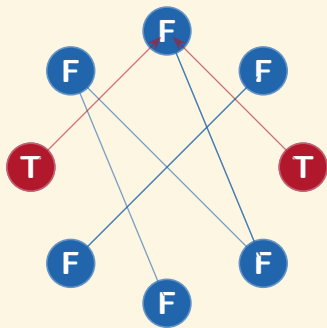


Traitor win probability by country (overall: 58.8%)





Random Voting



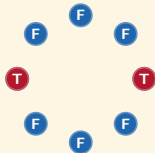
Proposition (Migdał¹ recurrence, adapted)

Let $w(n, m)$ denote the Traitor win probability under mutual random play from state (n, m) . Then

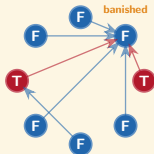
$$w(n, m) = \begin{cases} 0 & \text{if } m = 0, \\ 1 & \text{if } n \leq 2m, \\ \frac{n-m}{n} w(n-2, m) + \frac{m}{n} w(n-2, m-1) & \text{if } n > 2m. \end{cases}$$

¹Piotr Migdał. *A mathematical model of the Mafia game*. arXiv:1009.1031. 2010.

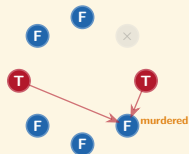
A

 $(n, m) = (8, 2)$ 

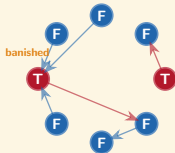
B

Banishment: $(n - 1, m)$ 

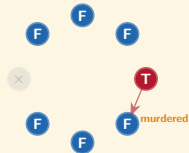
D

Murder: $(n - 2, m)$ 

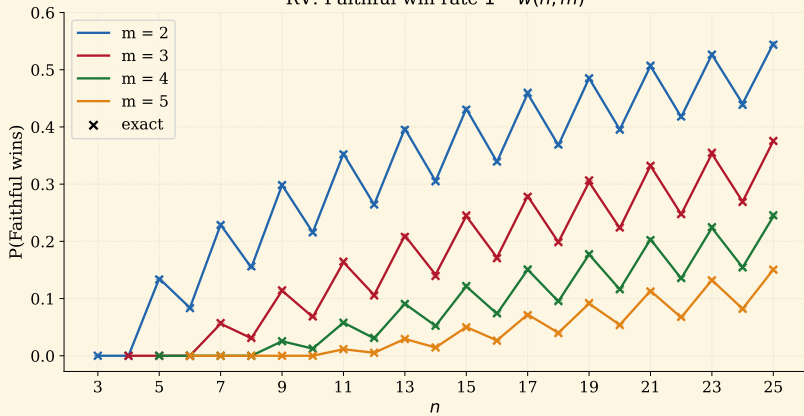
C

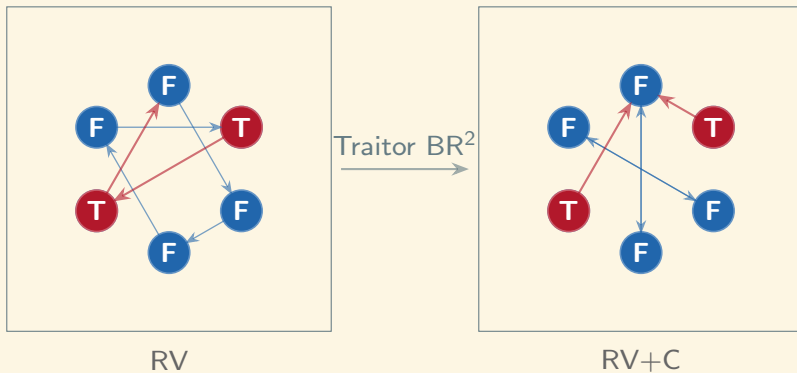
Banishment: $(n - 1, m - 1)$ 

E

Murder: $(n - 2, m - 1)$  $\frac{n-m}{n}$ $\frac{m}{n}$

RV: Faithful win rate $1 - w(n, m)$

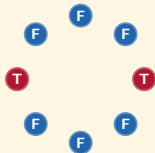




²Sizhe Wang. "Optimal strategy in the Werewolf game: A theoretical study". In: *Games: Research and Practice* (2024). arXiv:2408.17177.

A

$(n, m) = (8, 2)$

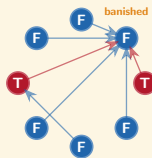


$p_F^\dagger(n, m)$

$1 - p_F^\dagger(n, m)$

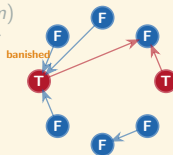
B

Banishment: $(n - 1, m)$



C

Banishment: $(n - 1, m - 1)$



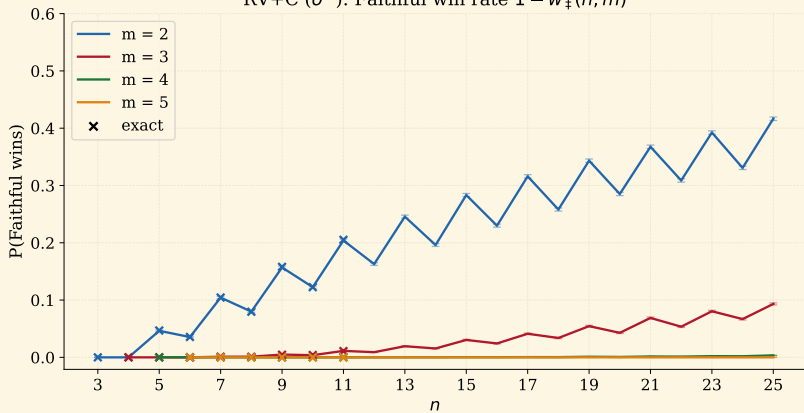
Lemma (Collusion banishment probability)

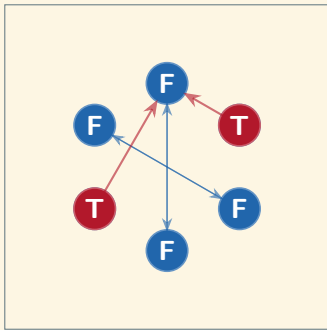
The probability that the eliminated player is Faithful under $\sigma^\dagger$ is

$$p_F^\dagger(n, m) = \frac{1}{|\Omega|} \sum_{v \in \Omega} \frac{|W(v) \cap F|}{|W(v)|},$$

where $W(v) = \{j : C_j(v) = M(v)\}$.

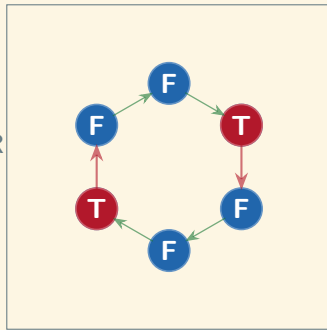
RV+C ($\sigma^\pm$): Faithful win rate $1 - w_+(n, m)$





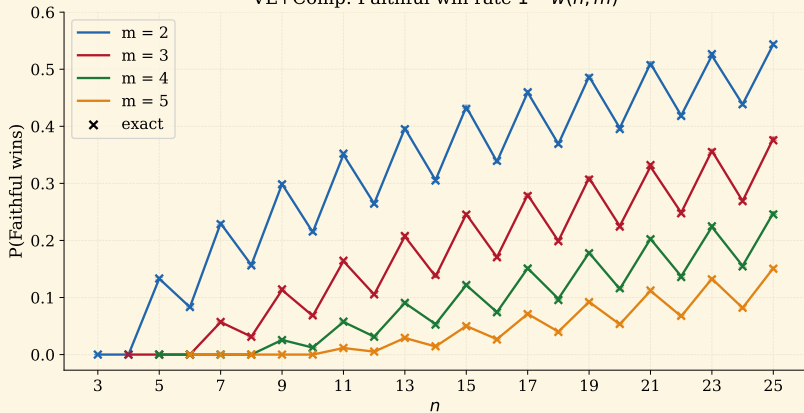
RV+C

Faithful BR



VL

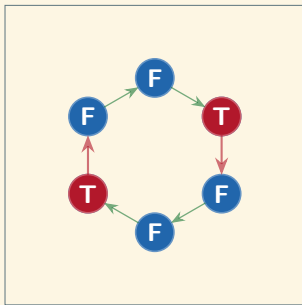
VL+Comp: Faithful win rate $1 - w(n, m)$



Theorem (Bayesian Nash Equilibrium (BNE))

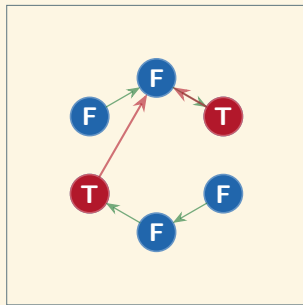
Vote-Left with Punishment constitutes a BNE of $TG(n, m)$ for every state with $n > 2m + 2$.

$\sigma^\dagger$ (VL+Opt): the Traitors' best response



$n > 2m + 2$: comply (VL)

late game



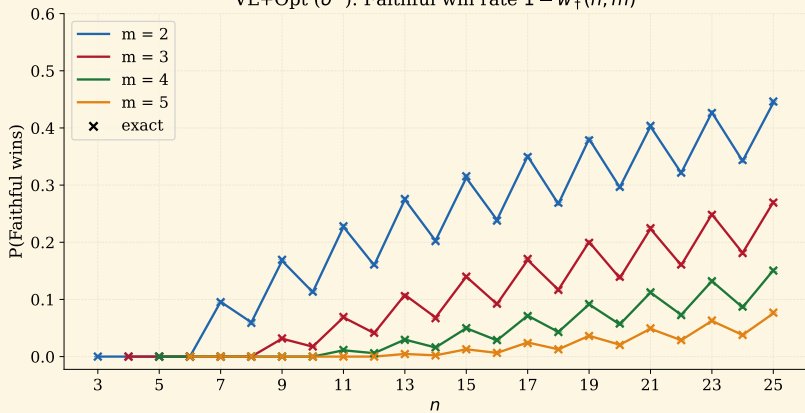
$n \leq 2m + 2$: collude

Proposition (Traitor win probability under $\sigma^\dagger$)

Let $w_\dagger(n, m)$ denote the Traitor win probability under Vote-Left with the optimal Traitor strategy $\sigma^\dagger$, starting from state (n, m) . Then

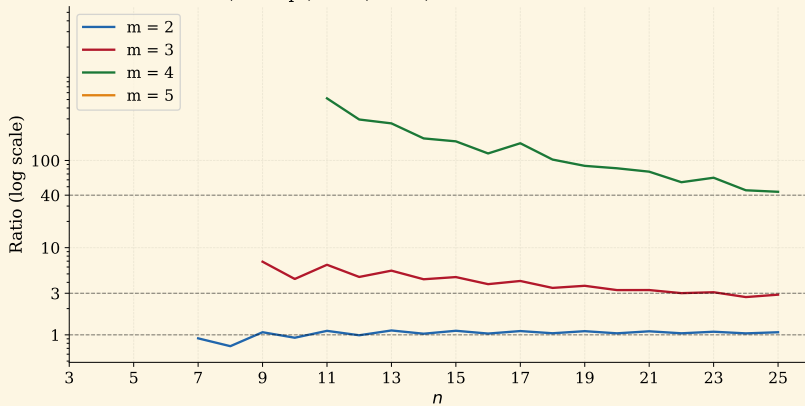
$$w_\dagger(n, m) = \begin{cases} 0 & m = 0, \\ 1 & n \leq 2m + 2, \\ \frac{n-m}{n} w_\dagger(n-2, m) + \frac{m}{n} w_\dagger(n-2, m-1) & n > 2m + 2. \end{cases}$$

VL+Opt ($\sigma^\dagger$): Faithful win rate $1 - w_+(n, m)$



(n, m)	w	RV	$w_{\ddagger}$	RV+C	$w_{\dagger}$	VL+Opt
(7, 2)	0.771	0.771	0.896	0.896	0.905	0.904
(8, 2)	0.844	0.846	0.920	0.920	0.941	0.940
(9, 3)	0.886	0.886	0.995	0.995	0.968	0.969
(10, 3)	0.931	0.932	0.996	0.996	0.982	0.983
(11, 3)	0.835	0.837	0.989	0.989	0.931	0.931
(11, 5)	0.988	0.989	1.000	1.000	1.000	1.000
(15, 2)	0.570	0.568	—	0.717	0.685	0.687
(20, 3)	0.776	0.777	—	0.957	0.860	0.863
(22, 3)	0.752	0.753	—	0.946	0.839	0.840
(24, 3)	0.731	0.731	—	0.933	0.819	0.819
(25, 3)	0.625	0.625	—	0.907	0.731	0.730
(22, 4)	0.864	0.865	—	0.999	0.927	0.927
(25, 4)	0.754	0.757	—	0.997	0.850	0.849

σ^+ (VL+Opt) / $\sigma^\pm$ (RV+C): ratio of Faithful win rates



Vote-Left raises the Faithful's winning probability

by a factor of approximately **three** at $m = 3$

by a factor of **forty or more** at $m = 4$

relative to random voting under Traitor collusion

The Vote-Left Equilibrium

Preprint: arxiv.org/abs/2605.10233

Install: `pip install backstab`

Docs: vknight.org/backstab

Dev: github.com/drvinceknight/backstab

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